

AK CONCERTS — MASTER SHEET GUIDE

PART 4: CI/CD Automated Site Build & Cloudflare Deployment

■ Overview

This technical guide details how the Astro static site generator executes prebuild data sync hooks (`scripts/update_events_from_sheet.js`), parses approved Master Sheet records, and automatically builds/deploys 3,500+ pages to Cloudflare Pages.

1. Build Pipeline Lifecycle

```
[ Trigger Build (Git Push / Webhook / Daily Cron) ]
      ▼
[ Prebuild Hook: update_events_from_sheet.js ]
  ■■■ 1. Check process.env.EVENTS_SHEET_URL
  ■■■ 2. Fetch CSV from Google Sheets
  ■■■ 3. Parse CSV rows & filter (status === 'Approved')
  ■■■ 4. Load local events.json
  ■■■ 5. Merge, deduplicate, and write to src/data/events.json
      ▼
[ Astro Build Hook: astro build ]
  ■■■ 1. Generate Static Pages (Cities, Venues, Bands, Events)
  ■■■ 2. Render JSON-LD Schemas (Event, MusicVenue, Breadcrumb, FAQ)
  ■■■ 3. Generate /sitemap-index.xml, /events.json, and /rss.xml
  ■■■ 4. Minify HTML & CSS
      ▼
[ Deployment to Cloudflare Pages CDN ]
```

2. Environment Variables Setup

Configure the following environment variable in Cloudflare Pages / GitHub Secrets:

```
EVENTS_SHEET_URL="https://docs.google.com/spreadsheets/d/e/2PACX-1v.../pub?output=csv"
```

3. Automated Webhook Deployment (Daily Auto-Refresh)

To trigger automatic daily website rebuilds when Cody approves new events:

1. In Cloudflare Pages Dashboard, navigate to **Settings → Builds & deployments → Deploy hooks**.
2. Click **Add deploy hook** (e.g. **Master Sheet Auto Sync**).

3. Copy the Webhook URL (e.g.

`https://api.cloudflare.com/client/v4/pages/webhooks/deploy_hooks/...`).

4. In Google Apps Script (**Code.gs**), add a daily trigger to ping the deploy hook whenever rows marked **Approved** are detected.

```
function triggerSiteRebuild() {  
  var webhookUrl = "https://api.cloudflare.com/client/v4/pages/webhooks/deploy_hooks/YOUR_HOOK_ID";  
  UrlFetchApp.fetch(webhookUrl, { method: "post" });  
}
```

4. Local Build & Verification Commands

To test sheet updates locally prior to committing:

```
# Set sheet URL in terminal environment  
export EVENTS_SHEET_URL="https://docs.google.com/spreadsheets/d/e/.../pub?output=csv"  
  
# Execute prebuild script & Astro static build  
npm run build  
  
# Preview build output locally  
npx astro preview
```